9	**(a)**		**DR** Using given values in $v = 1.2t^2 - kt^3$: $25.6 = 1.2\times 8^2 - k\times 8^3$ $k = \dfrac{25.6 - 76.8}{-8^3} = 0.1$	**M1** **A1** **[2]**	**3.3** **2.1**	method must be clear **AG**	Allow M1 for substitution of $t = 8, k = 0.1$ to obtain $v = 25.6$ A1 for comment made that $k = 0.1$ is correct value.
9	**(b)**		**DR** Uses $v = u + at$ $u = 0, v = 25.6, t = 8$ $\Rightarrow 25.6 = 0 + 8a$ $a = 3.2\ \text{m s}^{-2}$	**M1** **A1** **[2]**	**3.3** **1.1**	Cao	
9	**(c)**		**DR** First model $s = \int_0^8 (1.2t^2 - 0.1t^3)\,\text{d}t = \left[0.4t^3 - 0.025t^4\right]_0^8$ $(0.4\times 8^3 - 0.025\times 8^4) - (0.4\times 0 - 0.025\times 0)$ =102.4 m Second model: use *suvat* equation(s) with 3 from $u = 0, v = 25.6, t = 8, a =$ their value , giving e.g. $s = 0\times 8 + \frac{1}{2}\times 3.2\times 8^2$ or $s = \dfrac{1}{2}(0 + 25.6)8$ = 102.4 m, [which is the same value as for 1st model]	**M1** **M1** **A1** **M1** **A1** **[5]**	**3.4** **1.1** **1.1** **3.4** **1.1**	Attempt to integrate – expression for indefinite integral must be seen Both limits seen or establishes that c = 0 if indefinite integral used. Allow following first M mark Method must be clear Allow without comment	Allow SC1 for $\int_0^8 (1.2t^2 - 0.1t^3)\,\text{d}t = 102.4$